OpenFlow, Controllers, and Intents

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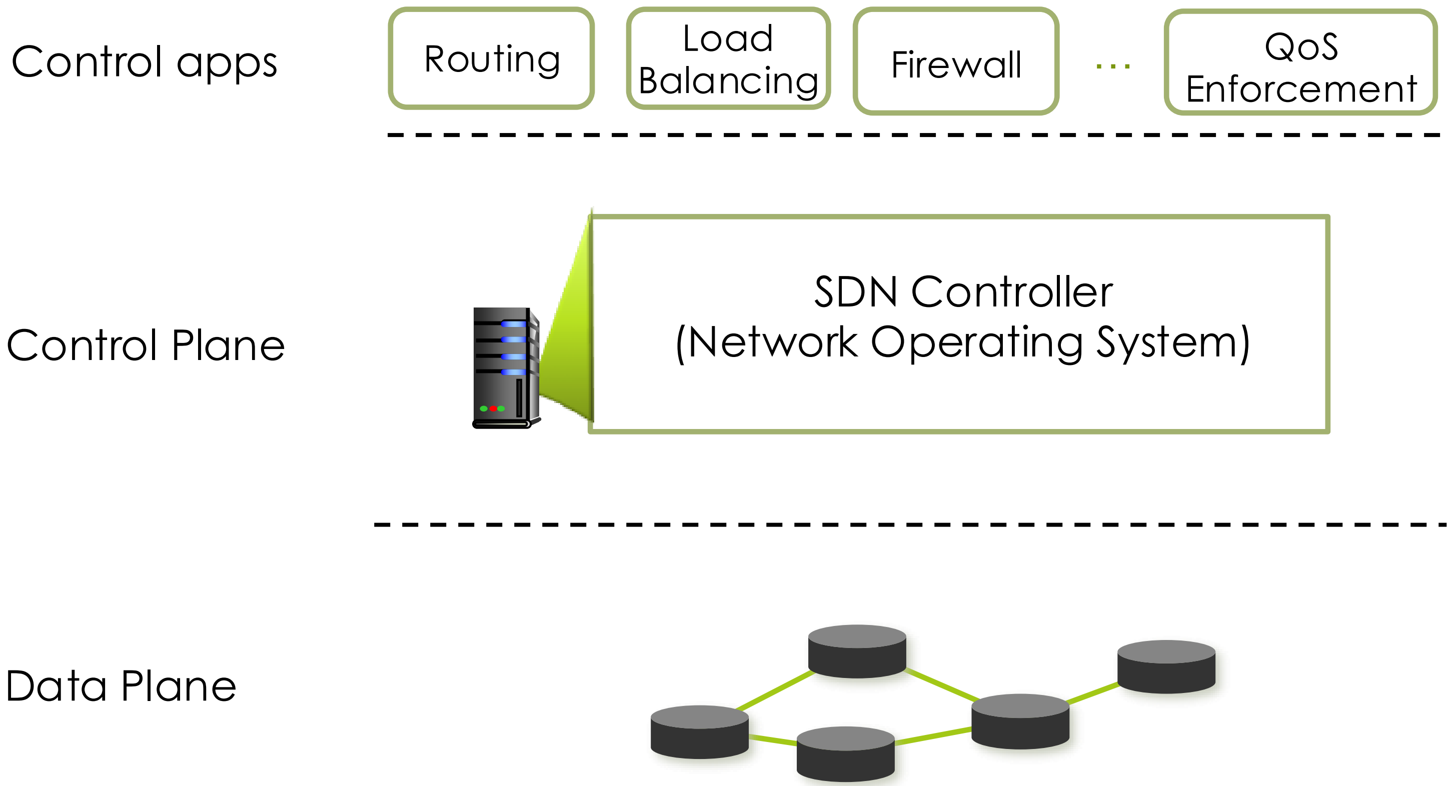
Rogers TEP Workshop 3: Transport Networks

Concepts #2

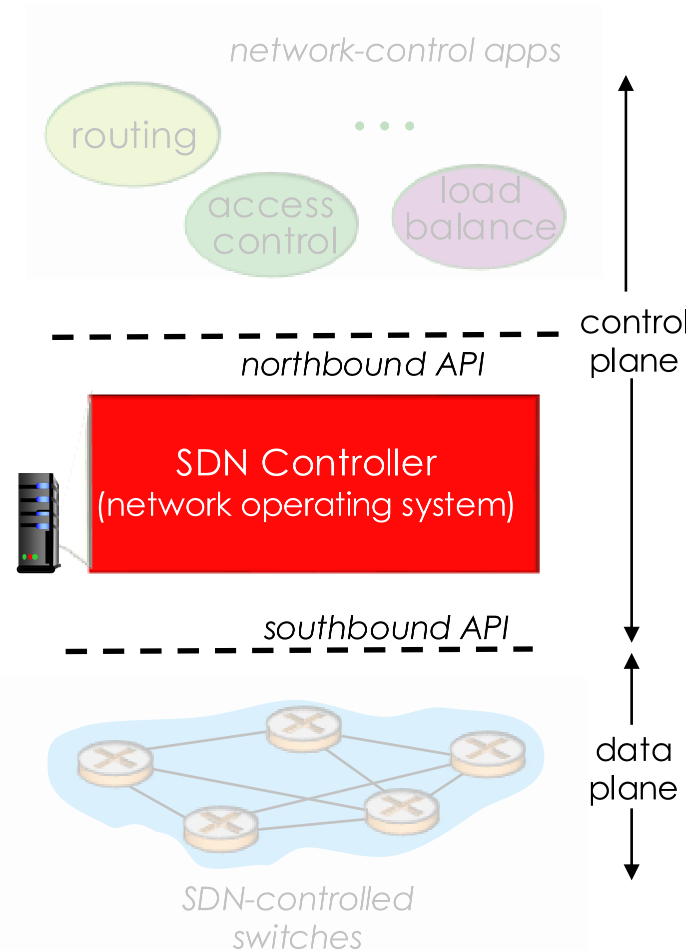
The SDN Control Plane

Overview of SDN planes

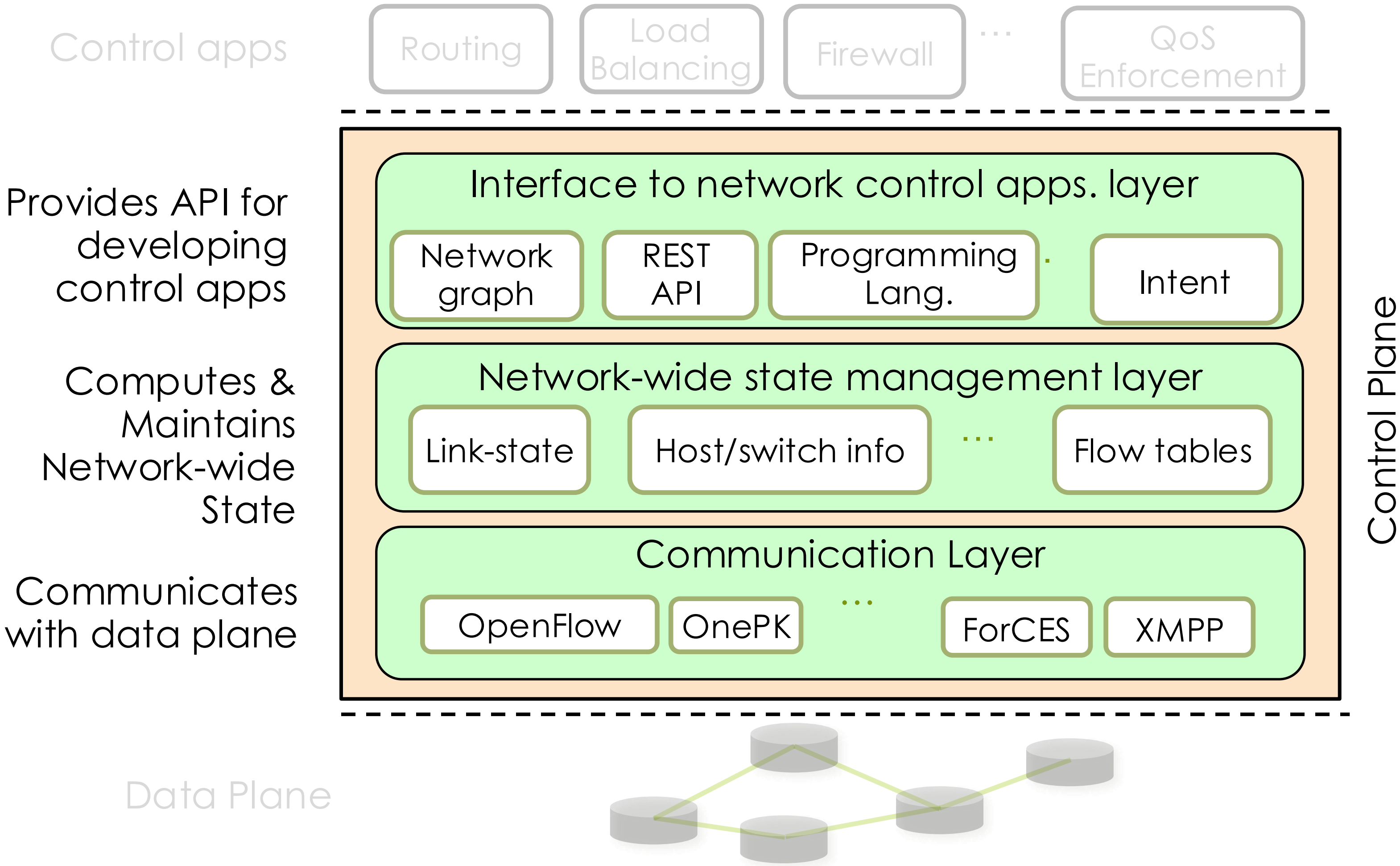


SDN Controller

- Network OS
  - logically centralized software to control data plane switches
  - maintain network state information
  - interacts with network control applications “above” via northbound API
  - interacts with network switches “below” via southbound APIs
    - supports one or more SB-APIs
- Controllers
  - OpenFlow: NOX/POX, Floodlight, OpenDaylight, ONOS, *etc*.
  - Other: Path Computation Element (PCE) in MPLS/GMPLS, ForCES, etc.

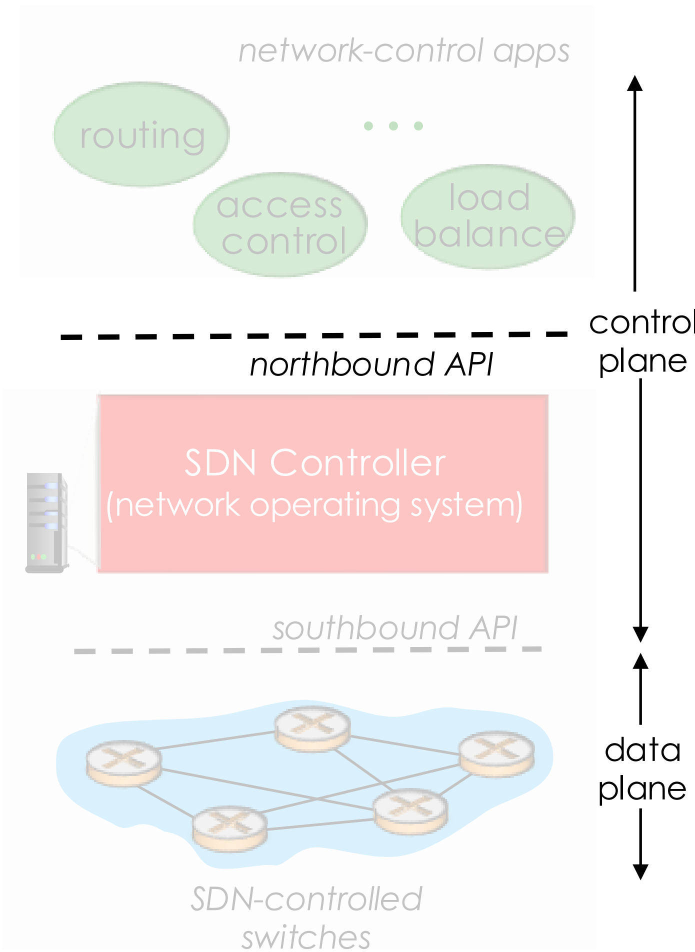


Components of SDN Controller



North Bound API

- Used by control apps to communicate with the controller
  - obtain network state
    - flow table content, counters, device health, *etc.*
  - modify network state
    - add/remove flow table entry, turn on/off a switch port, *etc.*
- Used by controller to provide network-wide view to control apps



What does a Controller do?

Control Plane Functions

- Topology Management
  - runs topology discovery protocol to discover network devices and their interconnections
- Path computation
  - uses topology information to compute shortest path and setup necessary forwarding rules for routing flows.
- Notification Management
  - receives, processes and forwards network event information to applications
- Statistics Collection
  - collects statistics from network devices by reading different counters
- Device Manager
  - configures network devices

Control Channel

- Each OpenFlow switch connects to the SDN controller over a TCP connection
- OpenFlow protocol defines a set of messages between the switch and the controller that allow:
  - Controller to program flow tables on the switches
  - Controller to query information from switches
  - Switches to send asynchronous notifications to the controller
- Control channel can be
  - In-band: Switches have default rules to forward control packets
  - Out-of-band: Control network can be a non-SDN network

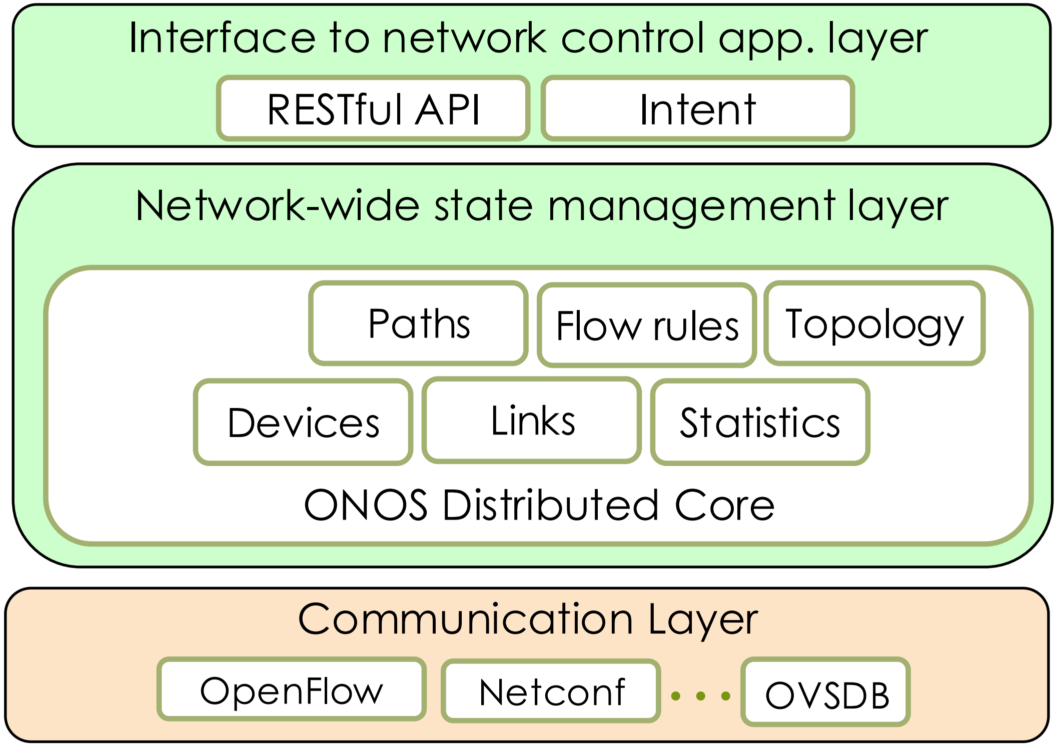
OpenFlow key messages

- Switch-to-controller messages
  - *PacketIn*: When a packet does not match any flow table entry the switch encapsulates the packet in a PacketIn message and sends it to the controller. This message is a request from the switch to the controller for path setup
  - *FlowRemoved*: When a flow table entry expires a FlowRemoved message is sent to the controller
  - *PortStatus*: A switch uses this message to inform controller about any status change of a port (*e.g.*, port went down)

The ONOS Controller

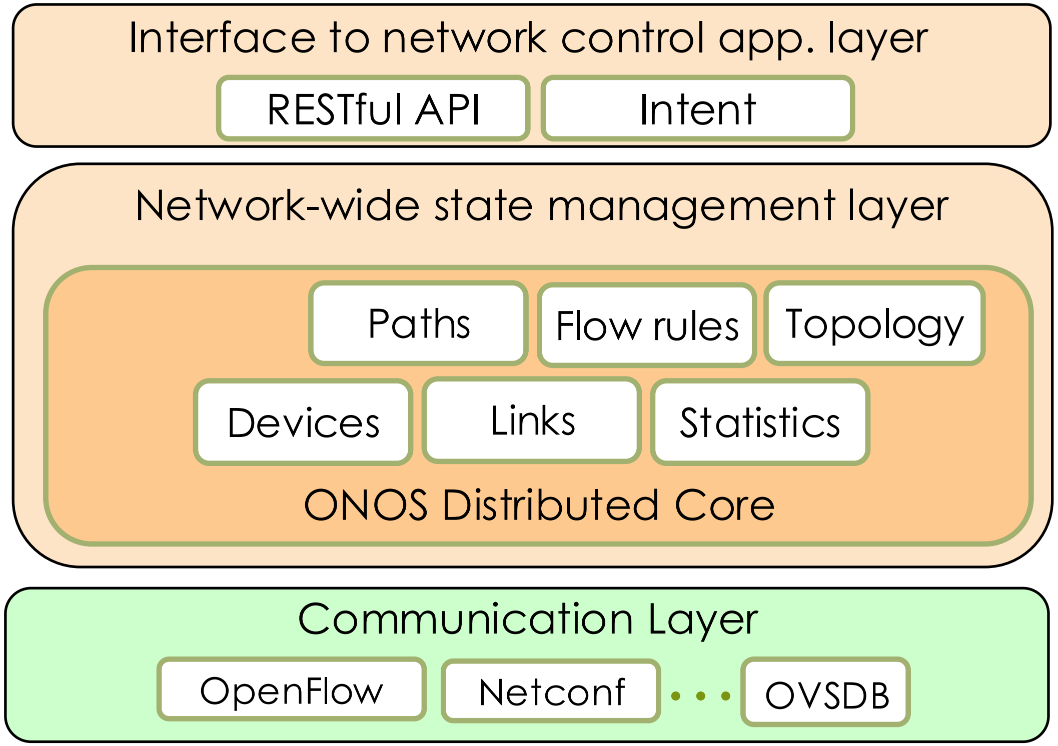
ONOS Controller

- designed as a distributed controller
- Communication layer
  - mostly supports OpenFlow
  - limited support for other southbound APIs



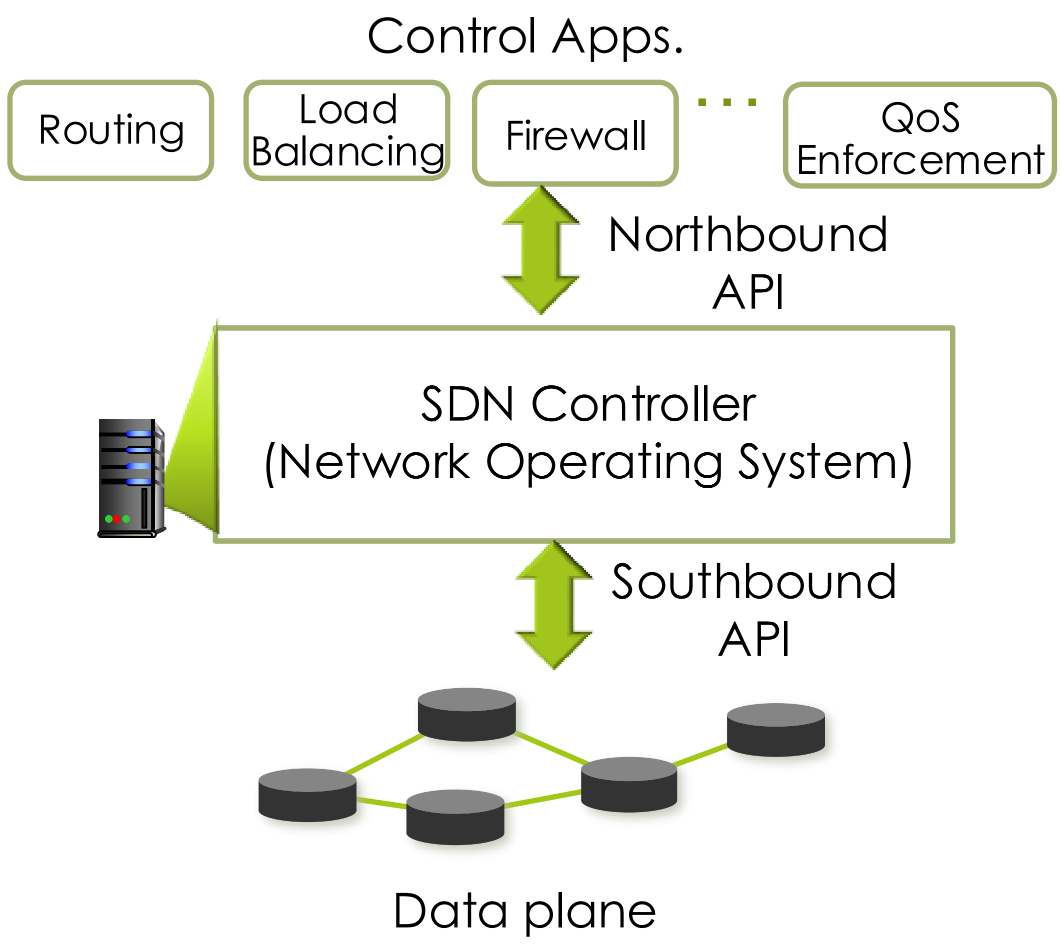
ONOS Controller

- ONOS ships with some basic network services
- Control apps external to controller
- Intent framework
  - high-level specification of service
  - *what* rather than *how*
- Distributed Core
  - service reliability
  - replication
  - performance scaling



SDN API Overview

- **Northbound API** is for communication between control plane and control apps
  - No standards yet
- **Southbound API** facilitates communication between SDN control and data plane
  - OpenFlow is the *defacto* standard



Northbound API

- Northbound API exists in many forms
  - Programming Language
  - Intent Language
  - RESTful API/Low level sockets programming
  - …

Intent

- Intent: Policy based derivatives describing "*what*" needs to be performed rather than "*how*" it needs to be performed

What:
Ensure connectivity between switch A and switch B

How:
- Setup flow between switch A and B
- Continuously monitor all switches on the path
- If something fails re-compute and reinstall flow rules

- Intent abstracts low level complexity from control applications
- Control applications specify their desired behavior through an intent framework supported by the controller
- Controller compiles intents to low level operations (e.g., flow installation) and takes the necessary actions

Example: ONOS Intent Framework

- allows control applications to express intent by specifying different constraints on network resources
- has two major components
  - Intent Compiler: Translates intents to installable actions on the network
  - Intent Coordinator: Determines how the network will be programmed/monitored to implement the intent

Traffic Engineering

- Traffic Engineering (TE) is done for different objectives
  - Maximizing network utilization
  - Ensuring QoS
  - Load balancing
  - Minimizing power consumption
- SDN is a very good fit for TE as
  - the controller maintains the state of the entire network
  - the optimal path for a traffic flow can be determined at a central point
  - traffic can be steered through specific paths i.e., the basis of transport slicing

From Theory to Practice

- You now have the full picture
  - Data plane: OVS switches forwarding packets
  - Control plane: ONOS managing the network automatically
  - Northbound API: REST interface to query and program ONOS
  - Intents: high-level connectivity expressed as "what, not how”
- In Lab 2 you take control of ONOS programmatically
  - Connect Mininet topology to ONOS
  - Query topology via REST API in Python
  - Install host intents: let ONOS handle the flow rules
  - Observe what happens when a link fails